

Adiba Yaseen Curriculum Vitae

PhD in Computer Science

AI and Machine Learning Researcher and University Lecturer

Riyadh Saudi Arabia | +966 59 089 9335 | adibayaseen@gmail.com

github.com/adibayaseen

Professional Profile

Computer scientist, AI and machine learning researcher, and university lecturer based in Riyadh. Holds a PhD in Computer Science with a final CGPA of 3.92 out of 4.00 and nearly five years of higher education teaching experience. Research applies machine learning, deep learning, graph neural networks, protein and molecular representations, and rigorous model evaluation to computational biology and AI driven drug discovery. Published work spans peptide hemolytic activity, compound protein interactions, protein binding affinity, and sequence based affinity prediction. Prepared for lecturer, assistant professor, AI research, data science, bioinformatics, computational biology, and drug discovery research roles in Saudi Arabia.

Core Expertise

- Artificial intelligence, machine learning, deep learning, graph neural networks, support vector machines, support vector regression, kernel methods, classification, regression, representation learning, and model benchmarking
- Computational biology, bioinformatics, protein protein interactions, compound protein interactions, peptide toxicity, binding affinity, molecular informatics, and computational drug screening
- Nonredundant cross validation, external validation, negative sample design, screening and ranking metrics, feature engineering, and reproducible computational experiments
- University teaching in artificial intelligence, machine learning, programming, object oriented programming, databases, operating systems, networks, and information technology

Academic Appointments

Lecturer | Department of Computer Science and Information Technology The University of Lahore

Lahore Pakistan | November 2020 to July 2024

- Taught Artificial Intelligence, Machine Learning, Computer Programming, Introduction to Information Technology, Database Management Systems, and Operating Systems and Networks.
- Delivered computing concepts through lectures, programming practice, coding demonstrations, learning management systems, and blended learning activities.
- Supported undergraduate learning and contributed to departmental teaching, student guidance, and academic activities.

Lecturer | Department of Computer Science and Information Technology The University of Lahore

Lahore Pakistan | September 2017 to June 2018

- Taught Programming Fundamentals I and Object Oriented Programming in C plus plus, including course code CS 1012 and CS 2730 offerings.
- Prepared and delivered lectures, laboratory exercises, programming assessments, and student support for foundational computing courses.

Lecturer | Department of Computer Science Government Degree College for Women

Shahkot Pakistan | October 2015 to February 2016

- Taught computer science at intermediate level using the Punjab Curriculum and Textbook Board syllabus for Classes 11 and 12.

Education

PhD in Computer Science | Pakistan Institute of Engineering and Applied Sciences

June 2018 to December 2024 | CGPA 3.92 out of 4.00

Focus Biomedical applications and machine learning

- Advanced study included Machine Learning, Machine Learning in Cyber Security, Applied Bioinformatics, Computational Biomolecular Design, Computational Interactomics, Information Retrieval and Data Mining, and Quantum Programming.

Master of Science in Computer Science | Pakistan Institute of Engineering and Applied Sciences

2016 to 2017 | CGPA 3.62 out of 4.00

Focus Bioinformatics and machine learning

- Completed 47 credits of coursework and 12 credits of thesis research. Coursework included algorithms, artificial intelligence, pattern recognition, bioinformatics, medical image processing, machine vision, deep neural networks, databases, data warehousing, and machine learning in bioinformatics.

Bachelor of Science in Computer Engineering | University of Engineering and Technology Lahore

2011 to 2015 | CGPA 3.351 out of 4.00

Focus Image processing and computer engineering

- Coursework covered programming, object oriented programming, data structures and algorithms, computer architecture, operating systems, networks, databases, artificial intelligence, embedded systems, digital signal processing, industrial automation, and digital design.

Research Projects

HemoNet Peptide Hemolytic Activity Prediction

- Developed a neural network framework combining contextual SeqVec peptide embeddings with SMILES based ECFP molecular fingerprints for N and C terminal modifications, including support for both L and D amino acids.
- Evaluated the method through repeated five fold cross validation, 90 percent and 70 percent nonredundant cross validation, an independent external dataset, and 28 clinical antimicrobial peptides.
- Reported about 88 percent AUC ROC compared with about 73 percent for HemoPI; the nonredundant model reached 72 percent AUC ROC on the external dataset compared with 40 percent and 44 percent for HemoPI and HemoPred.
- Released Python code, datasets, trained model resources, clustering utilities, UMAP analysis, and reproducible evaluation scripts through GitHub.

Compound Protein Interaction Evaluation and Drug Screening

- Developed and evaluated the Kernel CPI approach and HKRCPI research software for interaction prediction, target compound screening, and drug repurposing.
- Studied training and test similarity, synthetic negative construction, experimentally verified negatives, and the alignment of evaluation metrics with real screening use.
- Used RFPP to show that the model retrieved at least one known interacting compound in the top 10 for 85 percent of test proteins in a nonredundant screening experiment.
- Scored about 908250 protein drug pairs using SuperDRUG2 and supported screening studies involving human ACE2 and SARS CoV 2 Spike proteins.

Targeted Protein Complex Inhibitor Prediction

- Built a graph neural network pipeline that combines protein complex structure with compound SMILES representations to predict small molecule inhibition of a specified protein protein interaction.
- Trained and validated on 714 inhibitors across 23 protein complexes; reported cross validation AUC ROC of 0.85 and external AUC ROC values of 0.82 and 0.78.
- Compared the GNN pipeline with SVM and GearNet based approaches and released notebooks, datasets, features, results, and trained model artifacts.

Protein Binding Affinity Prediction

- Developed supervised learning models for protein interaction and binding affinity prediction using structural and interfacial features from 135 three dimensional protein structures.
- Applied SVM and SVR methods to analyze robustness to conformational changes and the effect of structural features; presented the work at ISACIS 2017 and IBCAST 2018.

Peer Reviewed Publications

Yaseen, A., Abbasi, W. A., and Minhas, F. U. A. A. (2018). Protein binding affinity prediction using support vector regression and interfacial features. In 2018 15th International Bhurban Conference on Applied Sciences and Technology (IBCAST), 194-198. DOI 10.1109/IBCAST.2018.8312222. [Link](#)

- Abbasi, W. A., Yaseen, A., Hassan, F. U., Andleeb, S., and Minhas, F. U. A. A. (2020). ISLAND: in-silico proteins binding affinity prediction using sequence information. *BioData Mining*, 13, article 20. DOI 10.1186/s13040-020-00231-w. [Link](#)
- Yaseen, A., Gull, S., Akhtar, N., Amin, I., and Minhas, F. (2021). HemoNet: Predicting hemolytic activity of peptides with integrated feature learning. *Journal of Bioinformatics and Computational Biology*, 19(5), article 2150021. DOI 10.1142/S0219720021500219. [Link](#)
- Yaseen, A., Amin, I., Akhter, N., Ben-Hur, A., and Minhas, F. (2022). Insights into performance evaluation of compound-protein interaction prediction methods. *Bioinformatics*, 38(Supplement 2), ii75-ii81. DOI 10.1093/bioinformatics/btac496. [Link](#)

Preprints and Research Manuscripts

- Yaseen, A., Roy, S., Akhter, N., Ben-Hur, A., and Minhas, F. (2024). Predicting small-molecule inhibition of protein complexes. *bioRxiv*. DOI 10.1101/2024.08.23.609286. [Link](#) *Archived preprint and not peer reviewed*

Teaching Profile

Courses taught Artificial Intelligence, Machine Learning, Programming Fundamentals in C plus plus, Object Oriented Programming in C plus plus, Computer Programming, Introduction to Information Technology, Database Management Systems, Operating Systems and Networks, and intermediate level Computer Science.

Additional areas prepared to support Deep Learning, Pattern Recognition, Bioinformatics, Computational Biology, Data Mining, Computer Vision, Medical Image Processing, and research methods in machine learning.

Instructional practice Lecture and laboratory delivery, programming demonstrations, coding simulations, learning management systems, blended learning, student mentoring, and accessible explanation of technical concepts.

Technical Skills

Programming and scientific computing Python, C plus plus, C, C sharp, SQL, MATLAB, Jupyter Notebook, PyTorch, TensorFlow, RDKit, BioPython, and PyMOL.

Machine learning Deep neural networks, graph neural networks, convolutional neural networks, SVM, SVR, random forests, XGBoost, kernel methods, classification, regression, transfer learning, feature engineering, and representation learning.

Bioinformatics and molecular AI Protein sequence and structure analysis, protein protein interactions, compound protein interactions, peptide activity and toxicity, binding affinity, SMILES, ECFP and Morgan fingerprints, SeqVec, GearNet, molecular similarity, and computational drug screening.

Evaluation and data methods Stratified and nonredundant cross validation, external validation, similarity controlled dataset design, negative sample construction, AUC ROC, AUC PR, MCC, sensitivity, specificity, ranking metrics, UMAP, CD HIT, and reproducible benchmarking.

Additional systems experience Oracle, ASP.NET, MPI, Pthreads, GPU computing, assembly language, Proteus, and Xilinx tools.

Open Source Research

HemoNet github.com/adibayaseen/HemoNet

- Prediction code, datasets, pretrained model resources, repeated cross validation, nonredundant evaluation, BLAST baselines, clustering, and UMAP visualizations.

HKRCPI github.com/adibayaseen/HKRCPI

- Kernel CPI implementation, NR HCPI and BindingDB resources, RFPP analysis, SuperDRUG2 screening, pair scoring, and ACE2 and Spike supplementary results.

PPI Inhibitors github.com/adibayaseen/PPI-Inhibitors

- GNN, SVM, and GearNet workflows with protein complex and inhibitor data, engineered features, trained models, evaluation notebooks, and final results.

Selected Additional Projects

- Gesture recognition using Kinect Built C sharp computer vision components for motion detection, background removal, skeletal feature extraction, gesture recognition, visual effects, and an augmented Pearl Hunting game.

- Intrusion detection Explored a one class SVM with deep belief networks and GPU based deep learning using TensorFlow.
- Cancer classification Designed an ensemble classifier for microarray data using correlation analysis and feature selection.
- Face understanding Studied automatic feature representation with convolutional and deep neural networks using GPU computing.
- Online hospital management system Developed a database backed web application using ASP.NET and SQL.
- Medical image enhancement Applied spatial filters and histogram equalization using ImageJ and DICOM software.

Presentations and Academic Engagement

- Presented MS research at the International Symposium on Advances in Computer and Information Science 2017 at PIEAS Islamabad.
- Presented the protein binding affinity paper at the 15th International Bhurban Conference on Applied Sciences and Technology in January 2018.
- Served as a resource person for Deep Learning activities at PIEAS in September 2016.
- Participated in ISACIS 2016 and ISACIS 2017, Applications of Computational Biology at NIBGE in 2017, SLATE online learning management faculty development training in 2017, and the International Conference on Advancements in Computational Sciences in 2018.

Scholarships and Research Support

- Higher Education Commission sponsored NRPU funded PhD research project June 2018 to June 2022.
- PIEAS IT Endowment Fund Fellowship sponsored by the Higher Education Commission for the MS in Computer Science 2015 to 2017.
- Fully funded Pakistan Engineering Congress scholarship for the BS in Computer Engineering 2011 to 2015.
- Merit scholarship for intermediate education at Shahab College based on secondary school performance.

Academic Service

- Active member of the Pakistani Students Welfare Organization, the ACM student community, and Software Square Computer Science Society at UET Lahore.
- Contributed to faculty development, research workshops, academic conferences, and student centered teaching activities.